

EBN122 KLASTOETS/CLASS TEST 3 (30 OKT 2013)

Vraag 1 [3]

a) Die stroom deur 'n 10mH induktor is:

$$i(t) = 5 \cos(20t) \text{ mA}$$

Wat is die spanning v_L oor die induktor?

b) Die stroom deur 'n 100 μ F kapasitor is:

$$i(t) = 5 \cos(20t) \text{ mA}$$

Wat is die spanning v_C oor die kapasitor indien $v_C(0)=0V$?

c) Die lading deur 'n 1mF kapasitor is:

$$q(t) = 10 \cos(20t) \text{ mC}$$

Wat is die stroom i_C deur die kapasitor?

Question 1 [3]

a) The current through a 10mH inductor is:

$$i(t) = 5 \cos(20t) \text{ mA}$$

What is the voltage v_L over the inductor?

b) The current through a 100 μ F capacitor is:

$$i(t) = 5 \cos(20t) \text{ mA}$$

What is the voltage v_C over the capacitor if $v_C(0)=0V$?

c) The charge through a 1mF capacitor is:

$$q(t) = 10 \cos(20t) \text{ mC}$$

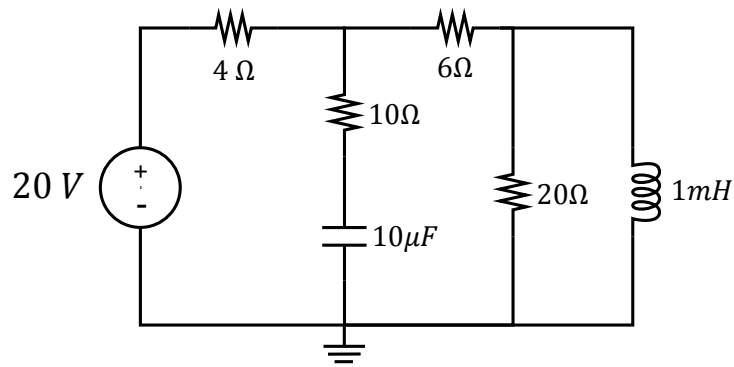
What is the current i_C through the capacitor?

Vraag 2 [2]

Bepaal die energie gestoor in die induktor en kapasitor respektiewelik.

Question 2 [2]

Determine the energy stored in the inductor and capacitor respectively.

**Vraag 3 [2]**

Bepaal die ekwivalente kapasitansie en ekwivalente induktansie tussen terminale *a* en *b* vir kringbane 1 en 2 onderskeidelik.

(Alle induktors het 'n waarde van 10mH.)

(Alle kapasitors het 'n waarde van 15μF.)

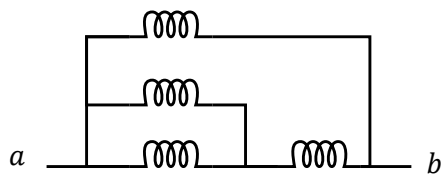
Question 3 [2]

Determine the equivalent capacitance and equivalent inductance between terminals *a* and *b* for circuits 1 and 2 respectively.

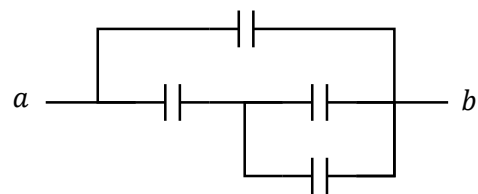
(All inductors have a value of 10mH.)

(All capacitors have a value of 15μF.)

Kringbaan / Circuit 1



Kringbaan / Circuit 2



Vraag 4 [2]

Gebruik fasors en skryf die volgende funksie as 'n enkele cosinus uitdrukking:

$$20\cos(\omega t + 60^\circ) - 10\sin(\omega t + 30^\circ)$$

Die antwoord moet in die volgende vorm wees:

$$A\cos(\omega t + \varphi), A > 0, -180^\circ \leq \varphi \leq 180^\circ$$

Question 4 [2]

Use phasors to write the following function as a single cosine expression:

$$20\cos(\omega t + 60^\circ) - 10\sin(\omega t + 30^\circ)$$

The answer must have the following form:

$$A\cos(\omega t + \varphi), A > 0, -180^\circ \leq \varphi \leq 180^\circ$$

Vraag 5 [2]

Watter sinusied loop voor en met hoeveel grade?

$$v_1(t) = 5\sin(\omega t - 130^\circ)$$

$$v_2(t) = 10\cos(\omega t - 10^\circ)$$

Question 5 [2]

Which sinusoid is leading and by how many degrees?

$$v_1(t) = 5\sin(\omega t - 130^\circ)$$

$$v_2(t) = 10\cos(\omega t - 10^\circ)$$

Vraag 6 [2]Bereken en skryf die antwoord in **polêre formaat**:

$$\frac{5e^{-j30^\circ} - 15 + j9}{-4 + j4}$$

Question 6 [2]Calculate and write the answer in **polar form**:

$$\frac{5e^{-j30^\circ} - 15 + j9}{-4 + j4}$$

Vraag 7 [2]

'n Sein met 'n hoekfrekwensie van $\omega = 20$ krad/s word aangewend oor 'n las met 'n impedanse van $Z = -j200 \Omega$. Bepaal of die las 'n kapasitor of 'n induktor is, en bepaal dan die waarde van die kapasitor of induktor.

Question 7 [2]

A signal with an angular frequency of $\omega = 20$ krad/s is applied across a load with an impedance of $Z = -j200 \Omega$. Determine if the load is an inductor or capacitor, and determine the value of the capacitor or inductor.